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(1) Research Program/Project/Study Title: Acquired Competencies, Least Mastered Topics and Performance of Freshmen College Students in Mathematics in the Modern World	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

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(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Mambusao						
		Province:	Capiz						
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(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
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(7) Sector/Commodity 		(8) Discipline/Field of Specialization Mathematics							

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

ABSTRACT

This study was conducted at Capiz State University Burias Campus this Second Semester, School Year 2022-2023 to determine the level of acquired competency, least mastered topics, and performance of freshmen college students in Mathematics in the Modern World. The respondents of the study were the 110 first year college students of the College of Education selected through simple random technique. This descriptive-correlational research made use of researcher-made and adopted questionnaire to collect the data. Consent to adopt the instrument was secured by the researcher before gathering the data. Data were analyzed using descriptive and inferential statistics and results were interpreted at 0.05 level of significance. Results of the study revealed that the first year college students had generally a “high” level of acquired competencies. The competencies in MMW are not only focused on acquiring knowledge but also in acquiring good values and skills which are significant in the achievement of the intended learning outcomes of the MMW. There are seven topics identified as the least mastered topics in the subject. These were “Logic, Negation, Connectives and Quantifiers”, and “Sets, Functions, Relations, and Binary Operations” in Chapter 2, Inductive and Deductive Reasoning, and Recreational Problems Using Math in Chapter 3, and “Measures of Dispersion” “Probability and Normal Distribution” and “Linear Regression and Correlation” in Chapter 4. Students had an average level of performance in the subject. The acquired competencies by the respondents were significantly related to their performance. This implies that the lesser the competency acquired by the students, the higher the probability of getting low performance.

Keywords: knowledge, values, skills, least mastered topics

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Mathematics in the Modern World (MMW) is one of the general education subjects taught in the college curriculum. It is a 3-unit course offered every first semester of the school year. There are four mandatory topics in this subject namely - Mathematics In Our World, Mathematical Language and Symbols, Problem Solving and Reasoning, and Data Management. There are two elective topics depending to a specific program or discipline depending on the need of the students to make up the whole course. The competencies in MMW are not only focused on the cognitive aspect but also affective and psychomotor skills which are significant in the achievement of the intended learning outcomes of the MMW.

According to Cabral (2022), the topics in MMW are very useful to students in their daily lives and by studying the topics it would help them to go beyond the typical interpretation of mathematics as simply a bunch of formulas, but as a source of aesthetics in patterns of nature.

In the study of Roman (2019), students experienced difficulties on the different topics of MMW with satisfactory performance in the subject. In addition, Nabayran (2022), identified the least mastered topics in MMW and found that mathematics learning in the new normal is challenging and difficult.

CAPSU Burias implemented modular learning during pandemic, and teachers were obliged to develop an instructional material in mathematics and other areas of the subject. Teachers make sure that these IMs are in accordance with the level of understanding of students and they provide more exercises and activities wherein students may acquire skills and knowledge from the lessons. They also make sure that the objectives and learning outcomes are met by the students.

Despite the efforts of the teachers to make students learn and acquire competence in the subject, challenges are still encountered by the students. The researcher believes that if students acquire the competencies intended for MMW, students can perform well in mathematics. Hence, this study.

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The objective of the study was to find out the acquired competency, least mastered topics and performance of students in Mathematics in the Modern World. Specifically, it sought answer to the following questions.

1. What is the level of competencies acquired by first year college students in Mathematics in the Modern World as a whole and when grouped according to sex and specialization?
2. What is the least mastered topics in Mathematics in the Modern World by the respondents?
3. What is the respondents' level of performance in Mathematics in the Modern World as a whole and when grouped according to profile variables?
4. Is the acquired competency by the respondents and their performance in MMW significantly related?

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

METHODOLOGY

This study employed a descriptive-correlational research. This study was conducted at CAPSU Burias Campus during the Second Semester of School Year 2022-2023. The respondents of the study were the 110 out of 154 first year college students of the College of Education. Cochran's formula was used to get the sample size. Then, the sample size was proportionately distributed to the different major/specialization using the formula for allocation. The data were gathered using researcher-made and adopted questionnaires. Consent to adopt the instrument was secured by the researcher before gathering the data. Data were analyzed using descriptive statistics such as mean, frequency and percentage and Spearman Rho for inferential statistics. Results were interpreted at 0.05 level of significance.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Profile of the Respondents

Sex. The result of the study revealed that freshmen college students is dominated by females (87.27%).

Specialization. Most of the respondents were BSED English (20.9%), followed by BSED Fil (17.3%), BEED and BPED (14.5%, respectively); and the least were BSED Math.

Table 1. Profile of the respondents

VARIABLES	FREQUENCY	PERCENT
Sex		
Male	14	12.7
Female	96	87.3
Total	110	100.0
Specialization		
BSED English	23	20.9
BSED Filipino	19	17.3
BSED Social Studies	14	12.7
BSED Science	14	12.7
BSED Math	8	7.3
BEED	16	14.5
BPED	16	14.5
Total	110	100.0

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Level of Acquired Competencies of First Year College Students in Mathematics in the Modern World

Findings of the study in Table 2 show that majority (60.9%) of the respondents had high level of competency acquisition. One-third (33.6%) of them had a very high level of acquisition and only few (5.5%) had an average level of competency acquisition. The mean of 4.01 generally indicates that respondents had high level of competency acquisition in the subject. The result of the study negates with the findings of Gloria (2015) that students have low mathematical competence.

In terms on the three indicators of competencies found in Table 3, students perceived that they had acquired high level of knowledge ($M=3.85$), values ($M=4.10$), and skills ($M=3.97$) in math. This result conformed with the findings of Roman (2019) that first year college students have high extent of acquisition on knowledge, values, and skills as set in the curriculum mapping of the Mathematics in the Modern World.

When respondents are grouped according to sex, both male ($M=3.94$) and female ($M=4.01$) have the same levels of acquisition as high level. When classified according to specialization, the level of acquisition of the respondents ranged from a very high (4.23) to a “high” level (3.68). BSED Math and BSED Science had a high level of acquired competency as compared to other specialization. (Table 4)

Table 2. Level of acquired competencies of the respondents.

VARIABLES	FREQUENCY	PERCENT
Very High	37	33.6
High	67	60.9
Average	6	5.5
Total	110	100.0
Mean=4.01, SD=0.41 (High)		

Table 3. Mean score on the level of acquired competencies per statement indicator

STATEMENT	Mean	SD	V.I.
KNOWLEDGE			
1. I can analyze "texts" (written, visual, oral, etc) critically.	4.25	0.59	Very High
2. I can demonstrate proficient and effective communication (writing, speaking and use of new technologies)	3.94	0.61	High
3. I can use basic concepts across the domains of knowledge.	3.91	0.61	High
4. I can demonstrate critical, analytical and creative writing.	3.73	0.66	High
5. I can apply different analytical models in problem solving.	3.43	0.70	High
Mean	3.85	0.48	High
VALUES			
6. I can appreciate the complexity of the human condition.	4.22	0.60	Very High
7. I can interpret the human experience from various perspective.	3.95	0.71	High
8. I can examine the purpose of communication from both Philippine and global perspectives.	3.95	0.67	High
9. I can take responsibility for knowing and being Filipino.	4.45	0.58	Very High

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10. I can reflect critically on shared contents.	3.95	0.63	High
11. I can generate innovative practices and solutions guided by ethical standards.	3.87	0.59	High
12. I can make decisions based on moral norms and imperatives.	4.05	0.60	High
13. I can appreciate various art forms.	4.23	0.67	Very High
14. I can contribute to aesthetics.	3.81	0.67	High
15. I can advocate respect for human rights.	4.39	0.61	Very High
16. I can contribute personally and meaningfully to the country's development.	4.18	0.61	High
Mean	4.10	0.41	High
SKILLS			
17. I can work effectively in a group.	4.19	0.60	High
18. I can apply computing tools to process information effectively.	3.81	0.68	High
19. I can use current technology to assist and facilitate learning and research.	4.14	0.71	High
20. I can negotiate the world of technology responsibly.	3.82	0.73	High
21. I can create solutions to problems in various fields.	3.78	0.70	High
22. I can manage one's knowledge skills and values for responsible and productive living.	4.03	0.64	High
23. I can organize one's self of lifelong learning.	4.05	0.71	High
Mean	3.97	0.50	High

Table 4. Level of acquired competencies of the respondent when grouped according to profile variables.

VARIABLES		MEAN	SD	V.I.
Sex				
	Male	3.94	0.38	High
	Female	4.02	0.41	High
Specialization				
	BSED English	4.09	0.31	High
	BSED Filipino	3.68	0.42	High
	BSED Social Studies	4.23	0.31	Very High
	BSED Science	4.22	0.36	Very High
	BSED Math	3.72	0.40	High
	BEED	3.99	0.35	High
	BPED	4.04	0.42	High

Least Mastered Topics in Mathematics in the Modern World

The least mastered topics in MMW can be found in figures 1-4. Results show that there are seven (7) topics which are not mastered by the first year college students. These topics were the following: In chapter 2 (Mathematical Language and Symbols), topics on “Logic, Negation, Connectives and Quantifiers”, and “Sets, Functions, Relations, and Binary Operations” shows 15.45%, and 35.64% freshmen students have correctly answered the items, respectively.

This was followed by Problem Solving and Reasoning in Chapter 3 where Inductive and Deductive Reasoning was at the bottom showing 11.82% of the respondents answer the items correctly, and Recreational Problems Using Math (33.64%).

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Furthermore, in Chapter 4 (Data Management), the bottom three topics were the “Measures of Dispersion” (15.00%), “Probability and Normal Distribution” (18.18%), and “Linear Regression and Correlation” (25.23%) where students got the correct answers, respectively.

The lowest percentages were considered as least mastered topics in MMW. The percentages were computed by dividing the number of students who got the test items on every topic per chapter correctly by total number of freshmen students.

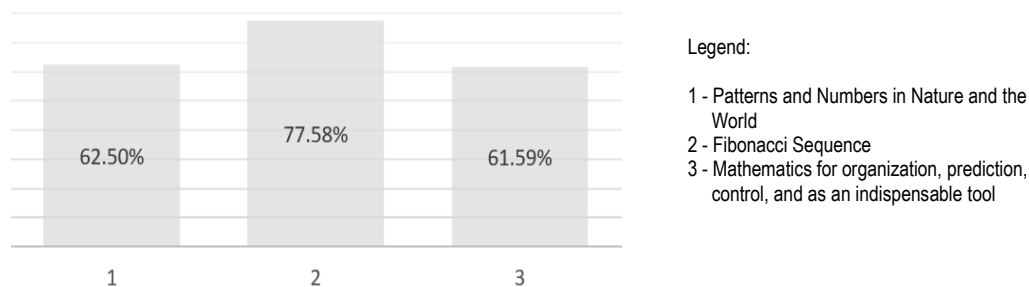


Figure 1. Percentage of Correct Responses of Freshmen Students per Topic in Chapter 1 – “Mathematics in Our World”

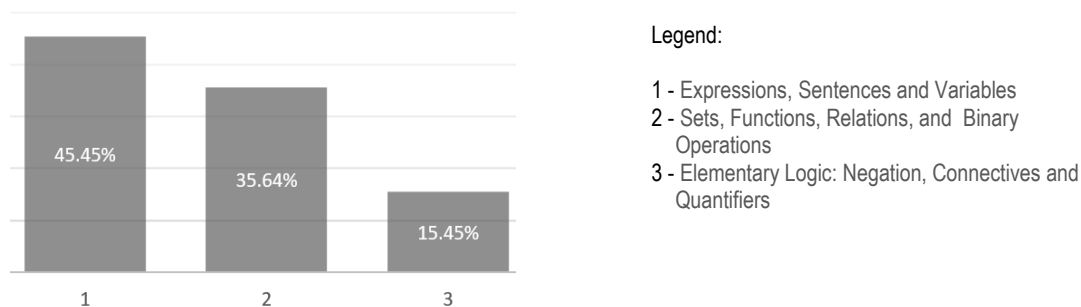


Figure 2. Percentage of Correct Responses of Freshmen Students per Topic in Chapter 2 – “Mathematical Language and Symbols”

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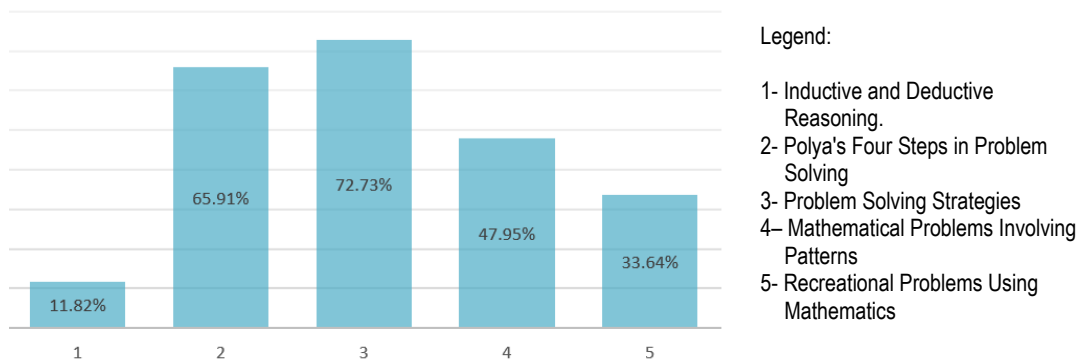


Figure 3. Percentage of Correct Responses of Freshmen Students per Topic in Chapter 3 – “Problem Solving and Reasoning”

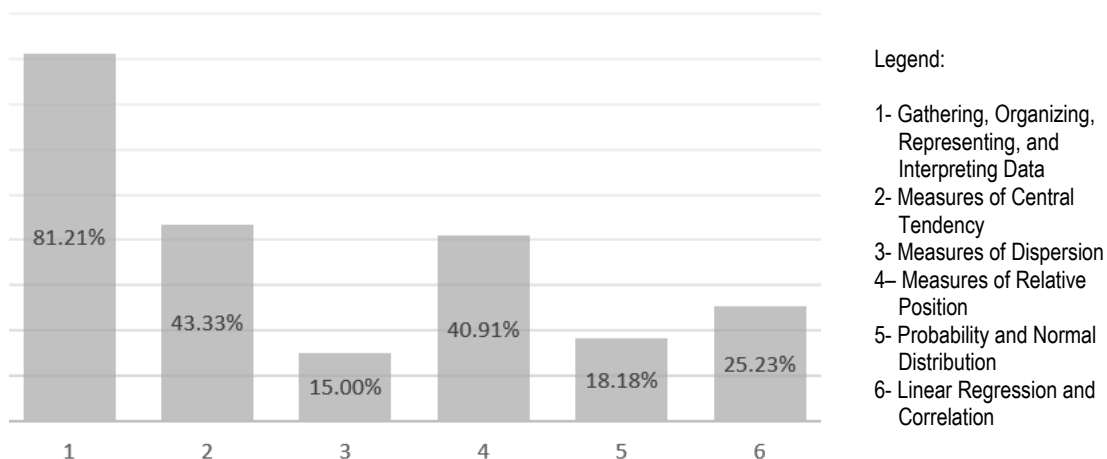


Figure 4. Percentage of Correct Responses of Freshmen Students per Topic in Chapter 4 – “Data Management”

Level of Performance of Students in Mathematics in the Modern World

Data in table 5 shows that majority (65.5%) of the students had an average level of performance while 30.0% of them had low performance in MMW. In general, the mean of 23.40 indicates that students had average performance in math.

When grouped according to profile variables (Table 6), data revealed that regardless of the respondents' sex, their performance level proved “average”, while when grouped according to specialization, BSED Filipino and BEED group found to had low performance compared to their counterparts.

Table 5. Level of performance of students in Mathematics in the Modern World.

VARIABLES	FREQUENCY	PERCENT
High	5	4.5
Average	72	65.5
Low	33	30.0

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Total **110** **100.0**
Mean=23.40, SD=5.12 (Ave)

Table 6. Level of performance of the respondent when grouped according to profile variables.

VARIABLES		MEAN	SD	V.I.
Sex				
	Male	24.07	5.73	Average
	Female	23.30	5.05	Average
Specialization				
	BSED English	26.91	3.44	Average
	BSED Filipino	20.21	4.96	Low
	BSED Social Studies	21.29	4.89	Average
	BSED Science	24.43	4.60	Average
	BSED Math	20.88	3.52	Average
	BEED	20.19	4.98	Low
	BPED	27.56	2.03	Average

Relationship between acquired competency by the respondents and their performance in MMW

Shown in Table 7 is the relationship between acquired competency by the respondents and their performance in MMW.

As disclosed, the r-value of 0.264 with 0.005 probability value is smaller than 0.05 alpha. Thus, results indicate a highly positive correlation between the two variables. The findings of the study leads to the rejection of the null hypothesis stating that these two variables are not significantly related. This affirms the study of Cabral (2022) that acquired competencies and difficulties in mathematics are significantly related to students' performance in MMW. It could be implied that when the respondents' acquired the competency required in MMW, their performance in math gets higher, however the lesser the acquired competency, the lower is their performance.

Table 7. Relationship between acquired competencies by the respondents and their performance.

VARIABLES	TEST	r-value	Sig
Acquired competency and Performance	Spearman rho	0.264**	0.005

** highly significant at 0.05 level

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSIONS

Based on the findings of the study, the following conclusions were made.

1. The first year college students have high level of acquired competence implying that students can analyze "texts" critically and can use current technology to assist and facilitate learning.

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2. The students have not mastered the topics namely Logic, Inductive and Deductive Reasoning, Measures of Dispersion, Probability and Normal Distribution, and Linear Regression and Correlation implying that students have difficulties in understanding the topics using modular approach and without the guidance of the professors.

3. The students taking MMW have average performance implying that there is a need of identifying factors which affect their performance in the said subject.

4. The acquired competency is related to performance. The more difficult for the student to cope up with the subject matter, the higher the probability of getting low performance This implies that the lesser the competency acquired, the lower is their performance in MMW and vice-versa.

RECOMMENDATION

1. Teachers should encourage students to acquire more knowledge, values, and skills in mathematics since this is the necessary requirements to make learning meaningful.

2. Teachers should provide more examples and activities and apply variety of teaching strategies and techniques to make students learn on the topics that are not mastered by the students. They would encourage students to participate and ask questions during class discussion since face to face classes are now implemented.

3. Since students have average performance in MMW, teachers may device peer tutorial classes for those students who have difficulty in mathematics.

4. Teachers should motivate and expose their students in real world to gain more experience and appreciate the beauty of Mathematics. With these, they would performed better in the subject.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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